Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Classical uncertainty principle

Files.

- File: `briefs/ClassicalUncertaintyPrinciple/ClassicalUncertaintyPrinciple.lean`
- Library: `ClassicalUncertaintyPrinciple`
- Imports: two new `ForMathlib` files:
  `ForMathlib/InformationTheory/GibbsInequality.lean`: Gibbs' inequality $-\int \rho \log \rho \le -\int \rho \log G$ for a probability density ρ and a positive G with $\int G \le 1$ (`MeasureTheory.integral_negMulLog_le`), with equality only for $\rho = G$ a.e. (`MeasureTheory.ae_eq_of_integral_negMulLog_eq`);
  `ForMathlib/Probability/Distributions/Gaussian/Real.lean`: the logarithm of the Gaussian density (`log_gaussianPDFReal`), and its mean and variance as Lebesgue integrals (`integral_mul_gaussianPDFReal`, `integral_sub_sq_mul_gaussianPDFReal`).

Setting. An ensemble on a measure space (X, μ) (`IsEnsemble` ρ μ) is a function $\rho \geq 0$ with $\int \rho \, d\mu = 1$ and $\rho \log \rho$ integrable (finite entropy). Its entropy relative to a cell of size c is `entropy` c ρ $\mu = -\int \rho \log(c\rho) \, d\mu$, in nats. A variable $f : X \to \mathbb{R}$ has mean `mean`, variance `var` and standard deviation `sd` under ρ. One DOF has phase space $\mathbb{R} \times \mathbb{R}$ with Lebesgue measure and cell $h > 0$, and $\hbar = h/2\pi$ (`hbar`). The conditions are:

- `LowerEntropicBound` h ρ (ZENT): $S_h[\rho] \geq 0$;
- `UncertaintyPrinciple` h ρ (CUP): $\sigma_q \sigma_p \geq \hbar/e$.

Finite second moments of the variables are hypotheses of the theorems. The Poisson bracket `poissonBracket` f g is $\partial_q f \, \partial_p g - \partial_p f \, \partial_q g$, with Fréchet derivatives.

Results.

Brief	Lean
Corollary 1, $S[\rho_U] = \log(\mu(U)/h^n)$	`entropy_uniform` (any measure space, any cell $c > 0$); `entropy_uniform_phaseSpace` (phase space $(\mathbb{R}^2)^n$, $c = h^n$)
Corollary 1, $S[\rho_U] = 0$ iff $\mu(U) = h^n$	`entropy_uniform_eq_zero_iff`
ZENT, CUP	`LowerEntropicBound`, `UncertaintyPrinciple`
Theorem 2, proof: a product of two independent Gaussians minimizes $\sigma_q \sigma_p$ at given entropy, and then $\sigma_q \sigma_p = (\hbar/e)e^S$	`hbar_div_exp_mul_exp_entropy_le`: $\sigma_q \sigma_p \geq (\hbar/e)e^{S_h[\rho]}$ for every ensemble; `sd_mul_sd_eq_iff`: equality iff ρ is a.e. a product of Gaussians
Theorem 2, eq. (3), ZENT implies CUP	`uncertaintyPrinciple_of_lowerEntropicBound`
CUP, equality case	`sd_mul_sd_eq_hbar_div_exp_iff`, corrected (see "Problems found")
Eq. (4), the Poisson bracket is the Jacobian determinant	`det_prod`; `poissonBracket_fst_snd`: $\{q, p\} = 1$
Theorem 2, eq. (5), $\sigma_f \sigma_g \geq (\hbar/e)e^{E[\log \lvert\{f,g\}\rvert]}$ under ZENT	`hbar_div_exp_mul_exp_le_sd_mul_sd`; for every ensemble $\sigma_f \sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log \lvert\{f,g\}\rvert]}$: `hbar_div_exp_mul_exp_entropy_add_le`
Remark, CUP does not imply ZENT	`exists_not_lowerEntropicBound`
(used for both equations) the Gaussian bound on any measure space	`Hyp.hbar_div_exp_mul_exp_le`: $\sigma_f \sigma_g \geq (\hbar/e)e^{S_h[\rho]+E[\log J]}$ for a weight $J > 0$ that turns every product of Gaussians in (f, g) into a subprobability density (`IsGaussWeight`)

Differences from the brief.

- The Lagrange-multiplier argument is replaced by Gibbs' inequality against the product G of Gaussians in q and p with the means and variances of ρ: $-\int \rho \log \rho \leq -\int \rho \log G = \log(2\pi e \, \sigma_q \sigma_p)$. This proves that the Gaussian is a minimum, not only a critical point, and gives the equality case. Only the marginal variances enter, so no covariance matrix is needed.
- Only one DOF is formalized: ZENT and CUP are stated for a density on $\mathbb{R}^2$ (the marginal of the DOF). Marginals of a density on $(\mathbb{R}^2)^n$ are not formalized; Corollary 1 is, for any n.

- The hypotheses that the brief leaves implicit are explicit: $\rho \geq 0$, $\int \rho = 1$, finite entropy and finite second moments. Positive variances are proved, not assumed: an ensemble of finite entropy cannot have zero variance in any variable, since Gibbs' inequality against ever narrower Gaussians would bound its entropy by $-\infty$.

- For eq. (5) the brief passes to the density ρ_{fg} of (f, g), with $\rho_{qp} = \rho_{fg} \, |\{f, g\}|$. Here ρ_{fg} is not formed. The Gaussian in (f, g) is pulled back to phase space with the weight $|\{f, g\}|$, and the change-of-variables formula shows that its integral is at most 1. This needs f, g differentiable, $(q, p) \mapsto (f, g)$ injective (not surjective, not orientation preserving), $\{f, g\} \neq 0$ everywhere, and $\rho \log |\{f, g\}|$ integrable. σ_f is the standard deviation of f under ρ. Physical units are not modelled, so the brief's assumption on them is not needed.

- The counterexample for the remark is a uniform distribution on two squares $[L, L + \varepsilon]^2$ and $[-L - \varepsilon, -L]^2$, with $\varepsilon = \sqrt{h}/2$ and $L = \sqrt{h}$. Its entropy is $\log \frac{1}{2} < 0$ and $\sigma_q \sigma_p \geq h/2 \geq \hbar/e$. The brief instead correlates q and p.

Problems found in the brief. The main claim (ZENT implies CUP, eqs. (3) and (5)) is correct; the equality case is misstated.

- CUP says that equality holds "if and only if position and momentum are independent gaussian distributions", and the proof ends the same way. Independent Gaussians satisfy $\sigma_q \sigma_p = (\hbar/e) e^S$, which equals $\hbar/e$ only when $S = 0$. Under ZENT, $\sigma_q \sigma_p = \hbar/e$ holds exactly for products of Gaussians with $\sigma_q \sigma_p = \hbar/e$ (`sd_mul_sd_eq_hbar_div_exp_iff`).

- The Lagrange multipliers give only a stationary point of $\sigma_q^2 \sigma_p^2$ at fixed entropy. They show neither that it is a minimum nor that a minimizer exists, which the proof then uses ("the gaussian is the distribution that minimizes the uncertainty").

- In the derivation of eq. (5), "order preserving" should be "orientation preserving"; the last two lines of the chain of equalities are identical; and CUP writes $E[\ln\{f, g\}]$ where the abstract and eq. (5) have $E[\ln |\{f, g\}|]$. The logarithm needs the absolute value when $\{f, g\} < 0$.

- The brief does not state that the entropy and the second moments must be finite. Without this $S[\rho]$ or σ may be undefined.